(Solution Bank–Final Answers: Mathematics For Computing)
Module 1: Vectors
Module 2: Matrices and Determinant
Module 3-Fundamental structures

Dr.Naheed Mohammed-Lecturer
PSC(Mathematics and Computing Unit)-UTAS Nizwa

1.1 Vectors and Linear Combinations

Example 1 **Final Answer:** The component form of the vector is $(3, 2)$, and the length is $\sqrt{13}$.
Example 2 **Final Answers:**

(i) $\vec{u} + \vec{v} = (4, 10)$

(ii) $\vec{u} - \vec{v} = (-10, -2)$

(iii) $\vec{v} - \vec{u} = (10, 2)$

(iv) $3\vec{u} = (-9, 12)$

(v) $2\vec{u} + 3\vec{v} = (15, 26)$

(vi) $|\vec{u}| = 5$

(vii) $|\vec{v}| = \sqrt{85}$

(viii) $|3\vec{u}| = 15$

Example 3 **Final Answer:** The component form is $(-3, 4, -2)$, and the length is $\sqrt{29}$.
Example 4 **Final Answers:**

(i) $\vec{u} + \vec{v} = (5, 4, 4)$

(ii) $\vec{u} - \vec{v} = (-1, -6, 2)$

(iii) $2\vec{u} + 3\vec{v} = (13, 13, 9)$

(iv) $|\vec{u}| = \sqrt{14}$

Example 5 **Final Answers:**

(i) $\vec{u} = 4\vec{i} + 9\vec{j}$

(ii) $\vec{v} = 3\vec{i} - 5\vec{j}$

(iii) $\vec{w} = 2\vec{i} + 5\vec{j} + 7\vec{k}$

(iv) $\vec{a} = 7\vec{i} - 2\vec{j} - 6\vec{k}$

Example 6 **Final Answer:** $\vec{u} + 2\vec{v} + 3\vec{w} = (14, 3, 8)$
Example 7 **Final Answer:** The vectors $(3, 4, 5)$ and $(6, 8, 10)$ are collinear.
Example 8 **Final Answer:** The vectors $\vec{u}$ and $\vec{v}$ are collinear.
Example 9 **Final Answer:** The vectors $\vec{u} = (0, 3, 1)$ and $\vec{v} = (0, 6, 2)$ are not collinear.
Example 10 **Final Answer:** The vectors $\vec{u} = (1, 2, 3)$ and $\vec{v} = (5, 10, 15)$ are collinear.
Example 11 **Final Answer:** $n = 6$

Exercise 1.1: Solutions

1). Classify the following measures as scalars and vectors

i) **Scalar**

ii) **Scalar**

iii) **Vector**

iv) **Vector**

v) **Scalar**

vi) **Scalar**

vii) **Scalar**

viii) **Scalar**

2). Find the component form and the length of the vector with initial point and terminal point are given below:

i) **Component form:** $(13, -1)$, **Length:** $\sqrt{170}$

ii) **Component form:** $(-3, -10)$, **Length:** $\sqrt{109}$

iii) **Component form:** $(9, -6, 3)$, **Length:** $\sqrt{126}$

iv) **Component form:** $(10, -1, -1)$, **Length:** $\sqrt{102}$

3). Find $\overrightarrow{u} + \overrightarrow{v}$, $\overrightarrow{u} - \overrightarrow{v}$, $2\overrightarrow{u} + 3\overrightarrow{v}$, $|\overrightarrow{u}|$, $|\overrightarrow{v}|$, $|3\overrightarrow{u}|$:

i) $\vec{u} + \vec{v} = (-3, 11)$, $\vec{u} - \vec{v} = (11, -7)$, $2\vec{u} + 3\vec{v} = (-13, 31)$,
Magnitudes: $|\vec{u}| = \sqrt{20}$, $|\vec{v}| = \sqrt{130}$, $|3\vec{u}| = 3\sqrt{20}$

ii) $\vec{u} + \vec{v} = (1, 4)$, $\vec{u} - \vec{v} = (-1, 2)$, $2\vec{u} + 3\vec{v} = (3, 9)$,
Magnitudes: $|\vec{u}| = 3$, $|\vec{v}| = \sqrt{2}$, $|3\vec{u}| = 9$

iii) $\vec{u} + \vec{v} = (11, 2, 15)$, $\vec{u} - \vec{v} = (-3, -8, -1)$, $2\vec{u} + 3\vec{v} = (29, 9, 38)$,
Magnitudes: $|\vec{u}| = \sqrt{74}$, $|\vec{v}| = \sqrt{138}$, $|3\vec{u}| = 3\sqrt{74}$

iv) $\vec{u} + \vec{v} = (-3, 7, 2)$, $\vec{u} - \vec{v} = (-1, -9, -2)$, $2\vec{u} + 3\vec{v} = (-7, 22, 6)$,
Magnitudes: $|\vec{u}| = \sqrt{5}$, $|\vec{v}| = \sqrt{69}$, $|3\vec{u}| = 3\sqrt{5}$

4). Write each of the following vectors as a linear combination of the two standard unit vectors.

i) $\vec{u} - 2 = \vec{i} + 6\vec{j}$

ii) $\vec{a} = 9\vec{i} + 12\vec{j}$

iii) $\vec{b} = 1\vec{i} - 5\vec{j} + 4\vec{k}$

iv) $\vec{w} = 8\vec{i} - 2\vec{j} - 3\vec{k}$

5). Find the linear combination of the following

i) $\vec{w} = (5, 1)$

ii) $\vec{w} = (-2, 3)$

iii) $\vec{w} = (1, 42, -60)$

iv) $2\vec{u} + 32\vec{v} - 5\vec{w} = (-5, -39, 54)$

6). Find x and y, where $(x, 3) = (2, x + y)$
$x = 2, y = 1$
7). Write the vector $\vec{v} = (1, -2, 5)$ as a linear combination of $\vec{u}_1 = (1, 1, 1), \vec{u}_2 = (1, 2, 3), \vec{u}_3 = (2, -1, 1)$
$\vec{v} = -6\vec{u}_1 + 3\vec{u}_2 + 2\vec{u}_3$
8). Are the vectors $(2, -3, 4)$ and $(-4, 6, -8)$ collinear?
Yes, the vectors are collinear.

1.2 Length and Dot Product

Example 1:
Find the dot product of the vectors $\vec{u} = (2, 5)$ and $\vec{v} = (4, 3)$.

Answer:
The dot product is $\boxed{23}$.

Example 2:
Find the dot product of the vectors $\vec{u} = (1, 3, 4)$ and $\vec{v} = (0, -1, 2)$.

Answer:
The dot product is $\boxed{5}$.
Example 3:
Find the angle between the vectors $\vec{u}$ and $\vec{v}$, given that $\vec{u} \cdot \vec{v} = \sqrt{3}$, $|\vec{u}| = 2$, and $|\vec{v}| = 1$.

Answer:
The angle is $\boxed{30^\circ}$.

Example 4:
Prove that the vectors $\vec{u} = (5, 4, 1)$ and $\vec{v} = (3, -4, 1)$ are orthogonal.

Answer:
The vectors are $\boxed{\text{orthogonal}}$.

Exercise 1.2

1. Find the dot product of the vectors and the lengths $|\vec{u}|$ and $|\vec{v}|$:
i) $\vec{u} = (4, 8), \vec{v} = (1, 5)$
Answer:
Dot product: $\boxed{44}$, $|\vec{u}| = 4\sqrt{5}$, $|\vec{v}| = \sqrt{26}$.

ii) $\vec{u} = (2, 3), \vec{v} = (7, 0)$
Answer:
Dot product: $\boxed{14}$, $|\vec{u}| = \sqrt{13}$, $|\vec{v}| = 7$.

iii) $\vec{u} = (2, -5, 6), \vec{v} = (8, 2, -3)$
Answer:
Dot product: $\boxed{-12}$, $|\vec{u}| = \sqrt{65}$, $|\vec{v}| = \sqrt{77}$.

iv) $\vec{u} = (1, -3, 4), \vec{v} = (3, 4, 7)$
Answer:
Dot product: $\boxed{19}$, $|\vec{u}| = \sqrt{26}$, $|\vec{v}| = \sqrt{74}$.

2. Find the angle between the vectors $\vec{u} = (1, 1, -1)$ and $\vec{v} = (2, 3, 4)$
Answer:
The angle is $\boxed{\theta = \cos^{-1}\left(\frac{1}{\sqrt{87}}\right)}$.

3. Find the value of k such that the vectors $\vec{u} = (2, k, 1)$ and $\vec{v} = (1, -2, 1)$ are perpendicular.
Answer: " The value of k is $\boxed{\frac{3}{2}}$.

2.1 Introduction to Matrices

Exercise 2.1: Solutions

1). Write the order of each of the following matrices:

a) The matrix $\begin{bmatrix} 2 & 5 & 4 \end{bmatrix}$ has order 1×3.

b) The matrix $\begin{bmatrix} 0 & 1 \\ 7 & 1 \\ 4 & 7 \end{bmatrix}$ has order 3×2.

c) The matrix $\begin{bmatrix} 7 & 1 & -2 \\ -3 & 2 & -7 \end{bmatrix}$ has order 2×3.

d) The matrix $\begin{bmatrix} 4 & 2 & 7 \\ 1 & 0 & 0 \\ 0 & 2 & 4 \end{bmatrix}$ has order 3×3.

2). Construct a 2×2 matrix, $A = [a_{ij}]_{2\times 2}$, whose elements are given by:

a)

$$A = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{5}{2} & 2 \end{bmatrix}$$

b)

$$A = \begin{bmatrix} 2 & \frac{9}{2} \\ \frac{9}{2} & 8 \end{bmatrix}$$

3). Construct a 3×4 matrix whose elements are given by $a_{ij} = \text{Max}\{i, j\}$:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 2 & 3 & 4 \\ 3 & 3 & 3 & 4 \end{bmatrix}$$

4). If a matrix has 18 elements, the possible orders are 1×18, 2×9, 3×6, 6×3, 9×2, and 18×1.

5). Three persons A, B, and C go to a store to buy stationery:

Matrix of purchase: $P = \begin{bmatrix} 15 & 10 & 5 \\ 10 & 8 & 4 \\ 12 & 7 & 4 \end{bmatrix}$ Matrix of prices: $Q = \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix}$

6). If $A = \begin{bmatrix} 6 & 7 & -3 \\ 3 & 2 & -4 \\ 1 & 0 & 2 \end{bmatrix}$, then $a_{23} + a_{13} = -7$.

2.2 Vectors and Linear Equations

Exercise 2.2: Solutions

1). Determine whether the following vectors are solutions of the linear equation $x + 2y - 4z = 3$:

a) For $\mathbf{u} = (3, 2, 1)$: **Yes, solution**.

b) For $\mathbf{u} = (1, 2, 4)$: **No, not a solution**.

2). Write the vector $\vec{\mathbf{v}} = (-8, 5)$ as a linear combination of the vectors $\vec{\mathbf{u}}_1 = (2, 3)$ and $\vec{\mathbf{u}}_2 = (-3, 4)$:

$$\vec{\mathbf{v}} = -1\vec{\mathbf{u}}_1 + 2\vec{\mathbf{u}}_2$$

Matrices: Basic Operations and Special Type of Matrices, Matrix Transpose

Exercise 2.3: Solutions

1). Find the values of x, y, and z from the following equations:
i) $x = 1$, $y = 4$, $z = 3$.
ii) $x = 2$, $y = 4$, $z = 0$.
iii) $x = 2$, $y = 4$, $z = 3$.

2). Let $A = \begin{bmatrix} -3 \\ 8 \\ 6 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 0 \\ 5 \end{bmatrix}$, and $C = \begin{bmatrix} 4 \\ 4 \\ -9 \end{bmatrix}$. Evaluate:

a)

$$A + B - C = \begin{bmatrix} -5 \\ 4 \\ 20 \end{bmatrix}$$

b)

$$6A - 5B = \begin{bmatrix} -28 \\ 48 \\ 11 \end{bmatrix}$$

3). Compute the following products:

i)

$$\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}$$

ii)

$$\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}$$

iii)

$$\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \times \begin{bmatrix} 1 & 3 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 6 & 12 \\ -1 & -3 & -6 \\ 3 & 9 & 18 \end{bmatrix}$$

iv)

$$\begin{bmatrix} 1 & 2 & 0 \\ 3 & 4 & -1 \\ 0 & 6 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & -1 & 2 \\ 2 & 2 & 4 \\ 1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 3 & 10 \\ 7 & 2 & 19 \\ 14 & 18 & 34 \end{bmatrix}$$

4). Can we multiply $[A]_{3\times 2}$ with $[A]_{3\times 3}$? **No**.
5). Find the order of AB, where $A_{5\times 2}$ and $B_{2\times 3}$.
The order of AB is 5×3.

6. Given

$$A = \begin{bmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix},$$

verify that:
a) $(A+B)^T = A^T + B^T$.
b) $(A-B)^T = A^T - B^T$.
c) $(5A)^T = 5A^T$.
7). Which of the following matrices are symmetric or skew-symmetric?
a) $\begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$ is **symmetric**.
b) $\begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is **skew-symmetric**.
c) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is **symmetric**.
d) $\begin{bmatrix} 2 & 16 \\ 16 & 4 \end{bmatrix}$ is **symmetric**.
e) $\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$ is **skew-symmetric**.
8). For the matrix

$$A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix},$$

verify that $A + A^T$ is a symmetric matrix:

$$A + A^T = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix}.$$

9). If

$$A = \begin{bmatrix} \sin x & \cos x \\ -\cos x & \sin x \end{bmatrix},$$

then $A^T A = I$:

$$A^T A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

2.4 The Determinant of a Matrix

Example 1: If $A = [7]$, then $|A| = 7$.
Final Answer: $|A| = 7$.
Example 2: If $A = \begin{bmatrix} 2 & 4 \\ -1 & 5 \end{bmatrix}$, then
Final Answer: $|A| = 14$.

Example 3: Find $|A|$, where $A = \begin{bmatrix} 1 & -2 & 4 \\ 1 & 1 & 13 \\ -2 & 6 & -1 \end{bmatrix}$.

Final Answer: $|A| = 3$.

Example 4: Find $|A|$, where $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$.

Final Answer: $|A| = -5$.

Example 5: Find the determinant for the matrix $\begin{bmatrix} -2 & 1 & 3 \\ 1 & 1 & 4 \\ -3 & 6 & 2 \end{bmatrix}$ using Sarrus' rule.

Final Answer: $|A| = 57$.

Remark:

Let A be an $n \times n$ matrix. It is said to be a singular matrix if $|A| = 0$.

Example 6: Show that the matrix $A = \begin{bmatrix} 2 & 1 & 5 \\ -3 & -1 & 4 \\ 6 & 2 & -8 \end{bmatrix}$ is singular.

Final Answer: $|A| = 0$ (singular matrix).

Example 7: Find the possible values of y, if $\begin{vmatrix} 1 & 2 & 0 \\ 1 & 2y & 3 \\ y & 2 & y \end{vmatrix} = 10$. **Final Answer:** $y = 2$ or $y = -4$.

Exercise 2.4: Solutions

1. Let $|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \lambda$, then find $|B| = \begin{vmatrix} ka & kb \\ kc & kd \end{vmatrix}$ where k is a non-zero constant.

Final Answer: $|B| = k^2\lambda$.

2. Evaluate the determinants

a) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$

Final Answer: $|A| = -12$.

b) $\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix}$

Final Answer: $|A| = 0$.

c) $\begin{vmatrix} 1 & 1 & -2 \\ 2 & 1 & -3 \\ 5 & 4 & -9 \end{vmatrix}$

Final Answer: $|A| = 0$.

d) $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix}$

Final Answer: $|A| = 18$.

e) $\begin{vmatrix} x-1 & x \\ x+2 & 2x \end{vmatrix}$

Final Answer: $|A| = x^2 - 4x$.

f) $\begin{vmatrix} -9 & 2 \\ -1 & 6 \end{vmatrix}$

Final Answer: $|A| = -52$.

3. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

Final Answer: $|2A| = -24$, $4|A| = -24$.

4. Find values of x for which

a) $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$

Final Answer: $x = \pm\sqrt{8}$.

b) $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$

Final Answer: $x = \pm\sqrt{3}$.

5. If A is a 2×2 matrix such that $|A| = 5$, find $|4A|$

Final Answer: $|4A| = 80$.

6. Find x, if $\begin{vmatrix} x & 5 \\ -2 & 4 \end{vmatrix} + 8 = 22$

Final Answer: $x = 1$.

2.5 Inverse Matrices : Solutions

Example 1: Find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 4 \\ -3 & 1 \end{bmatrix}$$

Final Answer:

$$A^{-1} = \begin{bmatrix} \frac{1}{14} & -\frac{2}{7} \\ \frac{3}{14} & \frac{1}{7} \end{bmatrix}$$

Example 2: Find, if possible, the inverse of the matrix

$$A = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$$

Final Answer: The matrix does not have an inverse, since $|A| = 0$.

Exercise 2.5

1. Given $A = \begin{bmatrix} 3 & -2 \\ 6 & 4 \end{bmatrix}$. Show that $\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$.
   Final Answer: $\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$.

2. Show that any matrix of the form $\begin{bmatrix} 0 & k \\ \frac{1}{k} & 0 \end{bmatrix}, k \neq 0$ is its own inverse.
   Final Answer: The matrix is its own inverse.

3. Find the inverse of each of the matrices (if it exists):

   (a) $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$ **Final Answer:**
   $$A^{-1} = \begin{bmatrix} \frac{2}{13} & -\frac{5}{13} \\ \frac{3}{13} & -\frac{1}{13} \end{bmatrix}$$

   (b) $\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ **Final Answer:**
   $$A^{-1} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{bmatrix}$$

   (c) $\begin{bmatrix} 6 & 8 \\ 3 & 4 \end{bmatrix}$ **Final Answer:** The matrix does not have an inverse , since $|A| = 0$.

2.6 Solve a linear system by Cramer's Rule: Solutions

Example 1:

a) $\left.\begin{array}{r} x + y = 5 \\ x - y = 3 \end{array}\right\}$ **Final Answer:**
$$x = 4, \quad y = 1$$

b) $\left.\begin{array}{r} 2x + 3y = 8 \\ 4x + 6y = 16 \end{array}\right\}$ **Final Answer:** Infinite number of solutions.

c) $\left.\begin{array}{r} 2x + 3y = 8 \\ 4x + 6y = 15 \end{array}\right\}$ **Final Answer:** No solution.

Example 2:
$$5x + 2y = 4$$
$$7x + 3y = 5$$

Final Answer:
$$x = 2, \quad y = -3$$

Example 3:
$$x - y + z = 4$$
$$x + y + z = 2$$
$$x - y + 2z = 4$$

Final Answer:
$$x = 3, \quad y = -1, \quad z = 0$$

Exercise 2.6

1. Use Cramer's Rule to solve the following system of linear equations:

a)

$$\begin{aligned} 3x + y - 19 &= 0 \\ 3x - y - 23 &= 0 \end{aligned}$$

Final Answer:

$$x = 7, \quad y = -2$$

b)

$$\begin{aligned} 7x + 3y &= 11 \\ 2x + y &= 3 \end{aligned}$$

Final Answer:

$$x = 2, \quad y = -1$$

c)

$$\begin{aligned} 2x - y + 6z &= 5 \\ x + 5y - z &= 7 \\ 3x + y + 2z &= 2 \end{aligned}$$

Final Answer:

$$x = -0.930, \quad y = 1.877, \quad z = 1.456$$

d)

$$\begin{aligned} -x + 3y + 1 &= 0 \\ 2x - y - 4z - 2 &= 0 \\ y + 2z - 4 &= 0 \end{aligned}$$

Final Answer:

$$x = 4.429 \quad y = 1.143, \quad z = 1.429$$

2.7 Application of Matrices in Computing

Exercise 2.7

1) **Explain some of the uses of Matrices in computer graphics. Answer:**

- Matrices are used for transformations, such as translation, scaling, and rotation.
- They are essential for perspective projections.
- Matrices are also used in rendering 3D objects onto 2D screens.

2) **Find the coordinates of the image of** $(2,4)$ **under the translation** $(x,y) \to (x+3, y-1)$**.**
Answer:

$$(2+3, 4-1) = (5,3)$$

3) **Find the coordinates of the image of** $(1,6)$ **under the translation**

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \end{bmatrix} \text{ where } t_x = -1, t_y = 12$$

Answer:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix} + \begin{bmatrix} -1 \\ 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 18 \end{bmatrix}$$

Coordinates: $(0,18)$

4) **Translate the polygon with coordinates** $A(2,7), B(7,10), C(10,2)$ **by 3 units in the** X**-direction and 4 units in the** Y**-direction.**
Answer:

$$A' = (2+3, 7+4) = (5,11), \quad B' = (7+3, 10+4) = (10,14), \quad C' = (10+3, 2+4) = (13,6)$$

5) **A point** $(4,3)$ **is rotated counterclockwise by an angle of** $45°$**. Find the rotation matrix** R **and the resultant point.**
Answer:

$$R = \begin{bmatrix} \cos 45° & -\sin 45° \\ \sin 45° & \cos 45° \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

Resultant point:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = R \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{4\sqrt{2}-3\sqrt{2}}{2} \\ \frac{4\sqrt{2}+3\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{7\sqrt{2}}{2} \end{bmatrix}$$

Simplified coordinates:

$$\left[\frac{\sqrt{2}}{2}, \frac{7\sqrt{2}}{2}\right]$$

6) **Scale a polygon with coordinates** $A(2,5), B(7,10), C(10,2)$ **by 2 units in the** X**-direction and 3 units in the** Y**-direction.**
Answer:

$$A' = (2\times 2, 5\times 3) = (4,15), \quad B' = (7\times 2, 10\times 3) = (14,30), \quad C' = (10\times 2, 2\times 3) = (20,6)$$

Module 3-Fundamental structures

Exercise 3.1-Sets and Set Operations

1. Represent the following sets by using the Listing Method.
   a. $\{x : x \in R, x^2 = 1\} = \{-1, 1\}$
   b. $\{x : x \in Z^+, x < 12\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$
   c. $\{x : x \in N, x^2 < 100\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
   d. $\{x : x$ is a solution of the equation $x^2 + x - 2 = 0\} = \{1, -2\}$
   e. $\{x : x$ is a letter of the word 'priority'$\}$ = {p, r, i, o, t, y}

2. Use set builder notation to describe the nature of each of these sets.
   (i) $\{3, 6, 9, 12\} = \{x : x = 3n, n \in N, 1 \leq n \leq 4\}$
   (ii) $\{-3, -2, -1, 0, 1, 2, 3\} = \{x : x \in Z, -3 \leq x \leq 3\}$
   (iii) $\{1, 2, 3, 4, 6, 12\} = \{x : x$ is a divisor of $12\}$

3. Suppose that $A = \{2, 4, 6\}, Q = \{4, 6\}, B = \{2, 6\}$ and $C = \{4, 6, 8\}$. Determine which among these sets are subsets of the other sets (represent the answers in symbols).
   $Q \subseteq A, B \subseteq A, Q \subseteq C$.

4. $X = \{1, 2, 3\}$. Find the power set of X.
   The power set $P(X) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

5. Let $X = \{p, q, r, s, t\}$ and $Y = \{p, q, r, s, t, u, v, w\}$. Find
   a) $X \cup Y = \{p, q, r, s, t, u, v, w\}$
   b) $X \cap Y = \{p, q, r, s, t\}$
   c) $X - Y = \phi$
   d) $Y - X = \{u, v, w\}$

6. Let $U = \{1, 2, 3, 4, 5, 6, 7\}, \quad X = \{1, 2, 3\}, \quad Y = \{2, 3, 4, 6\}$ and $Z = \{3, 4, 5\}$
   a) Create a Venn diagram for the above sets. (Detailed solution provided at the end of the Exercise 3.2 for your reference)

   b) $(X \cup Y) \cap Z = \{3\}$
   c) $Z' = \{1, 2, 6, 7\}$
   d) $X - (Y \cap Z) = \{1, 2\}$
   e) $X \cup (Y - Z) = \{1, 2, 3, 4, 6\}$
   f) $X \oplus Y = \{1, 4, 6\}$

7. If $|X \cup Y| = 12, |X| = 9$ and $|Y| = 7$, find $|X \cap Y|$.
   Using the formula $|X \cup Y| = |X| + |Y| - |X \cap Y|$,
   we have $12 = 9 + 7 - |X \cap Y| \implies |X \cap Y| = 4$.

8. Find $Y \times X$ if $X = \{a, b, c\}$ and $Y = \{0, 3, 9\}$.
   $Y \times X = \{(0, a), (0, b), (0, c), (3, a), (3, b), (3, c), (9, a), (9, b), (9, c)\}$.

9. Find $X \times Y \times Z$, if $X = \{10, 20\}, Y = \{1, 11\}$ and $Z = \{3\}$.
   $X \times Y \times Z = \{(10, 1, 3), (10, 11, 3), (20, 1, 3), (20, 11, 3)\}$.

10. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ with $a_i = i$. Write the bit string that represents the following sets:
    (i) $\{1, 3, 6, 10\} = 1010010011$
    (ii) $\{2, 3, 4, 7, 8, 9\} = 0111111100$
    (iii) $\{3, 4, 10\} = 0011000001$

11. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, determine the sets to which each bit string represents.
(i) $0101111000 = \{2, 3, 4, 5, 6, 7\}$
(ii) $1111001111 = \{1, 2, 3, 4, 8, 9, 10\}$
(iii) $1000000001 = \{1, 10\}$
12. Let A, B be subsets of $U = \{1, 2, 3, 4, 5, 6\}$. Determine the sets $A \cup B, A \cap B, A - B, A \oplus B$ and $\bar{A}$.

Set position U	1	2	3	4	5	6
Bit String A	0	1	1	0	0	0
Bit String B	0	0	1	1	1	0

Let $A = \{2, 3\}$ and $B = \{3, 4, 5\}$

$$\begin{aligned} A \cup B &= \{2, 3, 4, 5\}, \\ A \cap B &= \{3\}, \\ A - B &= \{2\}, \\ A \oplus B &= \{2, 4, 5\}, \\ \bar{A} &= \{1, 4, 5, 6\}. \end{aligned}$$

13. In a school, there are 20 teachers who teach Mathematics or Physics. Of these, 12 teach Mathematics and 4 teach both Physics and Mathematics. How many teach Physics?
Let M be the set of Mathematics teachers, and P be the set of Physics teachers. Then

$$|P| = |M \cup P| - |M| + |M \cap P| = 20 - 12 + 4 = 12.$$

14. In a class of 35 students, 24 like to play cricket and 16 like to play football. Also, each student likes to play at least one of the two games. How many students like to play both cricket and football?
Let C be the set of students who like cricket, F be the set of students who like football. Then

$$|C \cup F| = |C| + |F| - |C \cap F|,$$

we have:

$$35 = 24 + 16 - |C \cap F| \implies |C \cap F| = 5.$$

15. Assume that out of 1000 students, 200 are enrolled in math classes, 300 are enrolled in physics classes, and 50 are enrolled in both. How many individuals are enrolled in at least one of these classes? How many individuals have decided not to enroll in any of these courses? How many pupils enroll only in math?

$$|M| = 200, \quad |P| = 300, \quad |M \cap P| = 50.$$

we find:

$$|M \cup P| = 200 + 300 - 50 = 450.$$

Thus, the number of individuals enrolled in at least one of these classes is 450.
To find how many individuals have decided not to enroll in any of these courses, we subtract the number of students enrolled in at least one class from the total number of students:

$$1000 - |M \cup P| = 1000 - 450 = 550.$$

$$|M| - |M \cap P| = 200 - 50 = 150.$$

Exercise 3.2-Sequences and Summations

1. Find the following terms from the sequence $\{a_n\}$, where $a_n = 2(-3)^n + (5)^n$
(a). $a_0 = 2(-3)^0 + (5)^0 = 2 + 1 = 3$
(b). $a_1 = 2(-3)^1 + (5)^1 = -6 + 5 = -1$
(c). $a_4 = 2(-3)^4 + (5)^4 = 2 \cdot 81 + 625 = 162 + 625 = 787$
(d). $a_5 = 2(-3)^5 + (5)^5 = -486 + 3125 = 2639$
2. Find the term a_8 of the sequence $\{a_n\}$ if a_n equals
(a). $2^{n-1} \Rightarrow a_8 = 2^{8-1} = 2^7 = 128$
(b). $7 \Rightarrow a_8 = 7$
(c). $1 + (-1)^n \Rightarrow a_8 = 1 + (-1)^8 = 1 + 1 = 2$
(d). $-(-2)^n \Rightarrow a_8 = -(-2)^8 = -256$
3. Determine the initial five terms of the sequence generated by each of these recurrence relations and initial conditions.
(a). $a_n = 6a_{n-1}, \quad a_0 = 2 \Rightarrow a_0 = 2, a_1 = 12, a_2 = 72, a_3 = 432, a_4 = 2592$
(b). $a_n = a_{n-1}^2, a_1 = 2 \Rightarrow a_1 = 2, a_2 = 4, a_3 = 16, a_4 = 256, a_5 = 65536$
(c). $a_n = a_{n-1} + a_{n-3}, a_0 = 1, a_1 = 2, a_2 = 0 \Rightarrow a_0 = 1, a_1 = 2, a_2 = 0, a_3 = 3, a_4 = 5$

4. Verify whether the given sequence $\{a_n\}$ is a solution of the given recurrence relation.

(a) Recurrence Relation: $a_n = -a_{n-1}$, with $a_0 = 5$
Verify that the sequence $a_n = (-1)^n 5$ is a solution of the recurrence relation.
Solution:
Plug $a_n = (-1)^n 5$ in the recurrence relation and verify.

$$\text{LHS} = a_n = (-1)^n 5$$

$$\begin{aligned}\text{RHS} = -a_{n-1} &= -\left((-1)^{n-1}5\right) \\ &= (-1)^n 5.\end{aligned}$$

Thus, LHS = RHS, and the sequence $a_n = (-1)^n 5$ is a solution.
(b) Recurrence Relation: $a_n = 2a_{n-1}$, with $a_0 = 1$
Verify that the sequence $a_n = 2^n$ is a solution of the recurrence relation.
Solution:
Plug $a_n = 2^n$ in the recurrence relation and verify.

$$\text{LHS} = a_n = 2^n$$

$$\text{RHS} = 2a_{n-1} = 2(2^{n-1}) = 2^n.$$

Thus, LHS = RHS, and the sequence $a_n = 2^n$ is a solution.

(c) Recurrence Relation: $a_n = 2a_{n-1} - a_{n-2}$, with $a_0 = 5, a_1 = 10$
Verify that the sequence $a_n = 5(n+1)$ is a solution of the recurrence relation.
Solution:
Plug $a_n = 5(n+1)$ in the recurrence relation and verify.

$$\text{LHS} = a_n = 5(n+1)$$

$$\text{RHS} = 2a_{n-1} - a_{n-2}$$
$$= 2(5n) - 5(n-1+1) = 10n - 5n = 5(n+1).$$

Thus, LHS = RHS, and the sequence $a_n = 5(n+1)$ is a solution.
(d) Recurrence Relation: $a_n = a_{n-1} + 3$, with $a_0 = 1$
Verify that the sequence $a_n = 1 + 3n$ is a solution of the recurrence relation.
Solution:
Plug $a_n = 1 + 3n$ in the recurrence relation and verify.

$$\text{LHS} = a_n = 1 + 3n$$

$$\text{RHS} = a_{n-1} + 3 = (1 + 3(n-1)) + 3 = 1 + 3n.$$

Thus, LHS = RHS, and the sequence $a_n = 1 + 3n$ is a solution.
(e) Recurrence Relation: $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$, with $a_0 = 2$, $a_1 = 1$
Verify that the sequence $a_n = -n + 2$ is a solution of the recurrence relation.
Solution:
Plug $a_n = -n + 2$ in the recurrence relation and verify.

$$\text{LHS} = a_n = -n + 2$$

$$\text{RHS} = a_{n-1} + 2a_{n-2} + 2n - 9$$
$$= (-(n-1) + 2) + 2(-(n-2) + 2) + 2n - 9$$
$$= -n + 2.$$

Thus, LHS = RHS, and the sequence $a_n = -n + 2$ is a solution.
(f) Recurrence Relation: $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$, with $a_0 = 9$, $a_1 = 15$
Verify that the sequence $a_n = 7(2^n) - n + 2$ is a solution of the recurrence relation.
Solution:
Plug $a_n = 7(2^n) - n + 2$ in the recurrence relation and verify.

$$\text{LHS} = a_n = 7(2^n) - n + 2$$

$$\text{RHS} = a_{n-1} + 2a_{n-2} + 2n - 9$$
$$= (7(2^{n-1}) - (n-1) + 2) + 2(7(2^{n-2}) - (n-2) + 2) + 2n - 9$$
$$= 7(2^n) - n + 2.$$

Thus, LHS = RHS, and the sequence $a_n = 7(2^n) - n + 2$ is a solution.

5. Evaluate

(a) $\sum_{j=0}^{8} 3 \cdot (2^j) = 3 \cdot \left(2^9 - 1\right) = 3 \cdot (511) = 1533$

(b) $\sum_{j=0}^{8} 2 \cdot (-3)^j = 2 \cdot \frac{(-3)^9 - 1}{-4} = \frac{2 \cdot (19683 - 1)}{-4} = -9840$

6. Find $\sum_{j=100}^{200} j$.

Using the formula for the sum of an arithmetic series:

$$\sum_{j=100}^{200} j = \frac{n}{2}(a + l) = \frac{101}{2}(100 + 200) = 15150.$$

7. Find $\sum_{k=99}^{200} k^3$.

Using the formula for the sum of cubes:

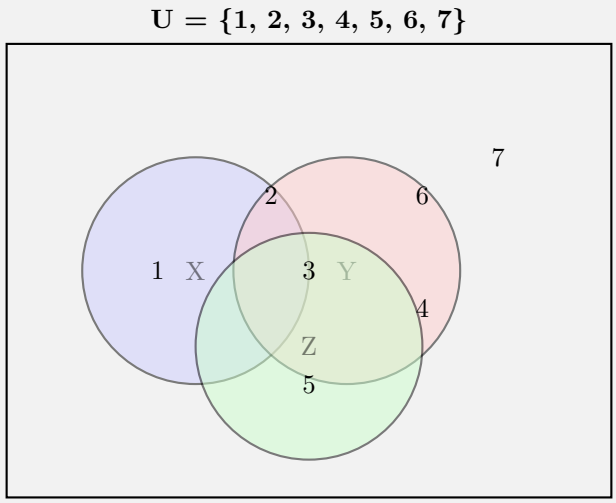
$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2 \implies \sum_{k=99}^{200} k^3 = \left(\frac{200 \cdot 201}{2}\right)^2 - \left(\frac{98 \cdot 99}{2}\right)^2 = 404010000 - 23532201 = 380477799.$$

8. Compute each of these double sums. (a) $\sum_{i=1}^{2} \sum_{j=1}^{3} (i + j) = 21$

(b) $\sum_{i=0}^{2} \sum_{j=1}^{3} (2i + 3j) = 72$

Venn Diagram for Sets

Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $X = \{1, 2, 3\}$, $Y = \{2, 3, 4, 6\}$, and $Z = \{3, 4, 5\}$.



Explanation

1. **Universal Set U**:
$$U = \{1, 2, 3, 4, 5, 6, 7\}$$
2. **Set X**:
$$X = \{1, 2, 3\}$$
3. **Set Y**:
$$Y = \{2, 3, 4, 6\}$$
4. **Set Z**:
$$Z = \{3, 4, 5\}$$
5. **Intersections**: - $X \cap Y = \{2, 3\}$ - $Y \cap Z = \{3, 4\}$ - $X \cap Z = \{3\}$ - $X \cap Y \cap Z = \{3\}$
6. **Elements Outside All Sets**: - The only element from U not in X, Y, or Z is 7.

Breakdown of the Venn Diagram Code

Universal Set Rectangle

- A rectangle is drawn around the sets to represent the universal set U.
- The label above the rectangle indicates the elements contained in $U = \{1, 2, 3, 4, 5, 6, 7\}$.

Circles for Sets

- Set X is drawn in blue, containing $\{1, 2, 3\}$.
- Set Y is drawn in red, containing $\{2, 3, 4, 6\}$.
- Set Z is drawn in green, containing $\{3, 4, 5\}$.
- The opacity is set to make the overlaps visible.

Element Placement

- Element 1 is only in X.
- Element 2 is in both X and Y.
- Element 3 is in all three sets X, Y, and Z.
- Element 4 is in both Y and Z.
- Element 5 is only in Z.
- Element 6 is only in Y.
- Element 7 is outside of all sets.